S4 — Biodiversity and Natural Resources

4.1 Classification and Taxonomy

The Hierarchy of Classification

Classification: The organisation of living organisms into groups based on shared features and evolutionary relationships. The hierarchical system arranges organisms from broad groups into increasingly specific categories.

The taxonomic hierarchy (from broadest to most specific):

1. **Domain** — Bacteria, Archaea, Eukarya (the three-domain system proposed by Carl Woese based on rRNA analysis)
2. **Kingdom** — e.g., Animalia, Plantae, Fungi, Protoctista, Prokaryotae
3. **Phylum** — e.g., Chordata (animals with a notochord/backbone)
4. **Class** — e.g., Mammalia
5. **Order** — e.g., Primates
6. **Family** — e.g., Hominidae
7. **Genus** — e.g., Homo
8. **Species** — e.g., sapiens

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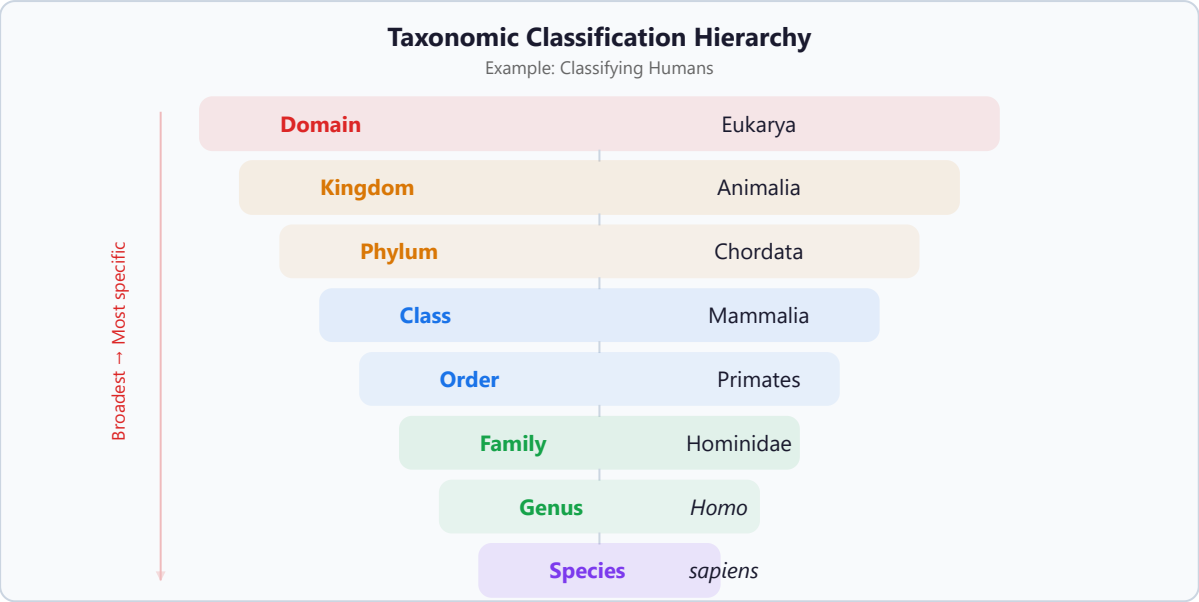
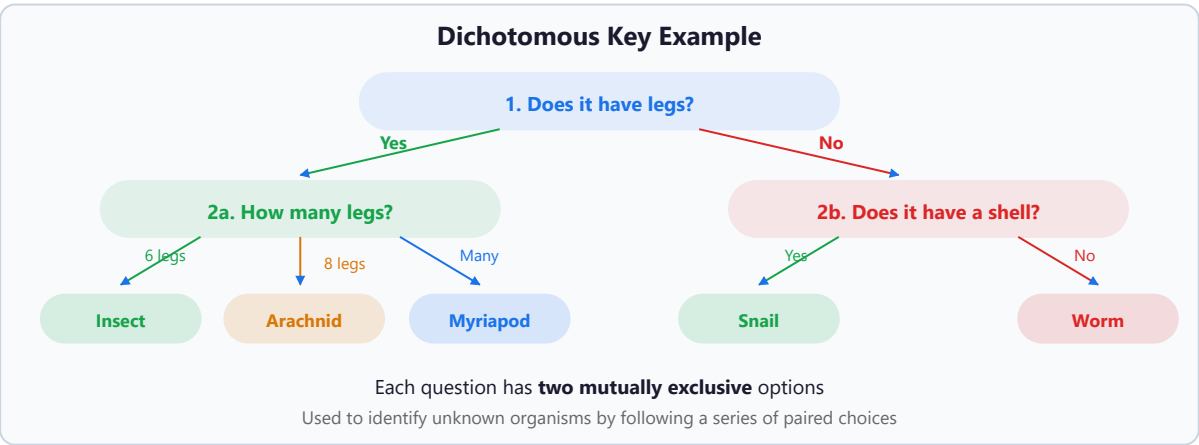


Species: A group of organisms that can interbreed to produce fertile offspring and are reproductively isolated from other groups. The biological species concept is widely used but has limitations (does not apply to asexual organisms, fossils, or organisms that hybridise).

Binomial Nomenclature

Every species is given a two-part Latin name (binomial): the genus name (capitalised) followed by the species name (lowercase), both written in italics (or underlined if handwritten). For example, *Homo sapiens*, *Escherichia coli*. This system was developed by Carl Linnaeus and provides a universal naming system understood by scientists worldwide regardless of language.



Phylogenetics

Modern classification increasingly relies on **molecular phylogenetics** — comparing DNA sequences, RNA sequences (particularly ribosomal RNA, which evolves slowly and is found in all organisms), or amino acid sequences to determine evolutionary relationships. Organisms that share more similar sequences diverged more recently and are more closely related. The number of differences between sequences can be used as a **molecular clock** to estimate the time since two species diverged, assuming a roughly constant mutation rate.

Phylogenetic trees (cladograms) show evolutionary relationships, with branch points (nodes) representing common ancestors and branch lengths sometimes indicating evolutionary distance. Molecular evidence has led to reclassification of many organisms, sometimes contradicting traditional morphological classification. For example, molecular evidence showed that fungi are more closely related to animals than to plants, and led Carl Woese to propose the three-domain system (Bacteria, Archaea, Eukarya) based on ribosomal RNA sequences, replacing the older five-kingdom system.

Advantages of molecular classification over morphological classification: It is more objective (based on quantifiable sequence data rather than subjective judgements about physical features); it can be applied to organisms with few observable features (e.g., bacteria); it reveals evolutionary relationships that convergent evolution may obscure (organisms may look similar due to similar environments, not shared ancestry); and it can resolve relationships at all taxonomic levels.

Exam Tip: Be prepared to interpret simple phylogenetic trees. The most closely related species share the most recent common ancestor (the nearest branch point). The length of branches may represent evolutionary time or the number of genetic changes. Remember that analogous structures (similar function, different evolutionary origin, e.g., wings of insects and birds) arise from convergent evolution and do not indicate close relationship, whereas homologous structures (same evolutionary origin, may have different functions, e.g., pentadactyl limb) indicate common ancestry.

4.2 Biodiversity

Levels of Biodiversity

Biodiversity exists at three levels:

- **Genetic diversity:** The variety of alleles within a species or population. Greater genetic diversity makes populations more resilient to environmental change.

- **Species diversity:** The number of different species (species richness) and the relative abundance of each species (species evenness) in an area.
- **Ecosystem diversity:** The variety of different ecosystems in a region (e.g., woodland, wetland, grassland, coral reef).

Measuring Biodiversity — Simpson's Diversity Index

Simpson's Diversity Index (D): A quantitative measure of biodiversity that takes into account both species richness and evenness. Values range from 0 to 1, where higher values indicate greater diversity.

$$D = 1 - \Sigma(n/N)^2$$

Where n = number of individuals of each species, and N = total number of all individuals. A community dominated by one species has low diversity (D close to 0), while a community with many equally abundant species has high diversity (D close to 1).

Exam Tip: When calculating Simpson's index, work through each species separately: divide each species count by total, square it, then sum all the squared values before subtracting from 1. Show your working clearly. Remember $D = 1 - \Sigma(n/N)^2$, not $\Sigma(n-1)/(N-1)$ — check which formula your exam board uses.

Sampling Techniques

To measure biodiversity in the field, sampling techniques must be used because it is usually impractical to count every organism:

- **Quadrats:** Square frames placed randomly to sample sessile organisms (plants, slow-moving invertebrates). Record species frequency, percentage cover, or density. Random sampling using coordinates from a random number generator reduces bias.
- **Transects:** A line or belt transect is used to study changes in community composition along an environmental gradient (e.g., from shore to inland).
- **Mark-release-recapture:** Used for mobile animals. Capture a sample, mark them, release, then recapture. Population size is estimated using the Lincoln index.

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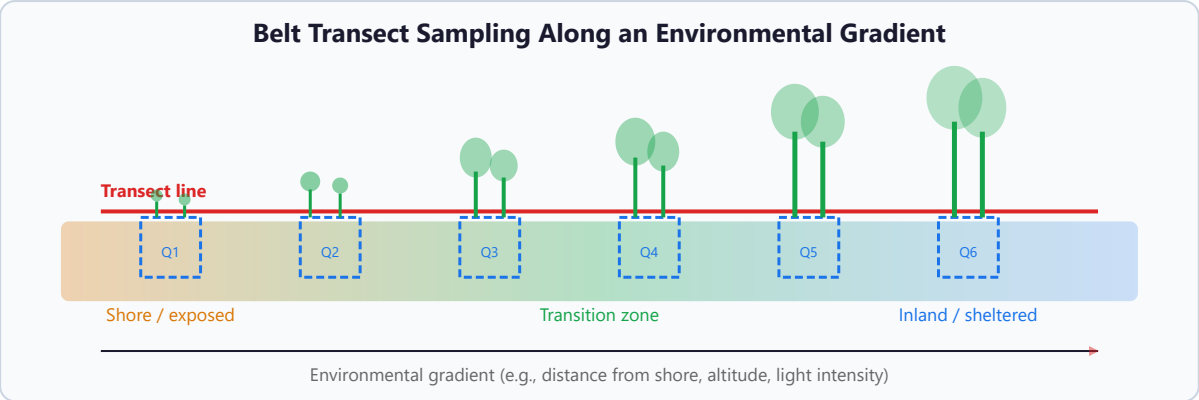

$$\text{Population estimate} = (\text{number in first capture} \times \text{number in second capture}) \,/\, \text{number of marked recaptures}$$

4.3 Plant Cell Biology

Plant Cell Structure

Plant cells share many features with animal cells (nucleus, mitochondria, ribosomes, rough and smooth ER, Golgi apparatus, cell membrane) but also have distinctive features:

- **Cell wall:** Made of cellulose microfibrils embedded in a matrix of hemicellulose and pectins. Provides structural support, prevents the cell from bursting (lysis) when turgid, and is fully permeable to water and dissolved substances.
- **Chloroplasts:** Double membrane-bound organelles containing thylakoid membranes (arranged in grana) and stroma. Site of photosynthesis. Contain their own DNA and 70S ribosomes.
- **Large permanent vacuole:** Membrane-bound (tonoplast), filled with cell sap (water, ions, sugars, pigments). Maintains turgor pressure, supports the cell, and stores substances.
- **Plasmodesmata:** Channels through cell walls connecting the cytoplasm of adjacent cells, allowing transport of molecules and communication between cells.
- **Amyloplasts:** Starch-storing plastids found in storage organs like potato tubers and root cells.

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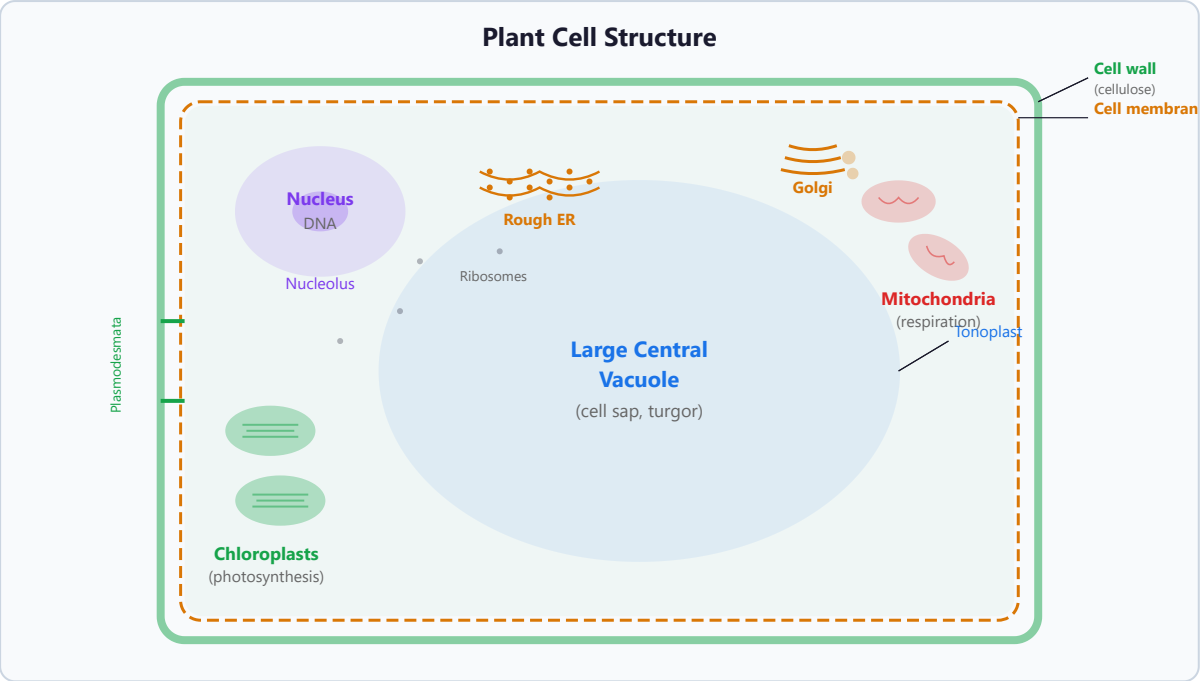

Cellulose and Plant Fibres

Cellulose is a polysaccharide made of long, unbranched chains of **beta-glucose** monomers joined by 1,4-glycosidic bonds. Because beta-glucose molecules are oriented alternately (each rotated 180° relative to its neighbour), the chain is straight rather than coiled. Adjacent chains are cross-linked by extensive **hydrogen bonds**, forming microfibrils of great tensile strength (comparable to steel on a weight-for-weight basis). These microfibrils are embedded in a matrix of hemicellulose and pectins, creating a composite material similar in principle to fibreglass or reinforced concrete.

Cellulose cannot be digested by most animals because they lack the enzyme **cellulase** — only certain bacteria, fungi, and protists produce it. Ruminant animals (e.g., cows) have mutualistic microorganisms in their rumen that produce cellulase, allowing them to break down plant cell walls. Cellulose is the most abundant organic molecule on Earth, with plants producing an estimated 10^{11} tonnes per year.

Comparing structural polysaccharides: Cellulose (in plant cell walls) and chitin (in fungal cell walls and arthropod exoskeletons) are both structural polysaccharides that are insoluble, strong, and chemically unreactive. Chitin is similar to cellulose but has an amino acid group added to each glucose monomer (N-acetylglucosamine), making it even stronger.

Plant fibres (sclerenchyma and xylem) are strengthened by additional compounds: lignin (waterproofs and adds rigidity) and sometimes suberin. These natural fibres have many human uses including textiles (cotton, flax/linen), rope (hemp, sisal), paper, and building materials.

4.4 Plant Uses and Sustainability

Plants as a Resource

Humans depend on plants for food (crops), medicines (e.g., aspirin from willow bark, digitalis from foxglove, taxol from yew), fibres, fuels (wood, biofuels), and building materials. Plant breeding and genetic modification have enhanced crop yields, nutritional content, and pest resistance. However, monoculture farming reduces biodiversity and increases vulnerability to disease.

Drug Discovery from Plants

Many pharmaceuticals are derived from plant secondary metabolites (chemicals produced for defence or signalling, not directly involved in growth). Examples include:

Drug	Plant Source	Medical Use
Aspirin (acetylsalicylic acid)	Willow bark (*Salix*)	Pain relief, anti-inflammatory, anticoagulant
Digitalis	Foxglove (*Digitalis purpurea*)	Treatment of heart failure (increases cardiac output)
Taxol (paclitaxel)	Pacific yew (*Taxus brevifolia*)	Anti-cancer drug (prevents cell division)
Quinine	Cinchona bark (*Cinchona*)	Anti-malarial
Morphine	Opium poppy (*Papaver somniferum*)	Pain relief (analgesic)

The destruction of biodiversity-rich habitats (such as tropical rainforests) may result in the loss of undiscovered medicinal compounds. This provides an economic argument for conservation alongside ecological and ethical arguments.

Sustainability

Sustainability: Meeting the needs of the present generation without compromising the ability of future generations to meet their own needs. In biological terms, this means using resources at a rate

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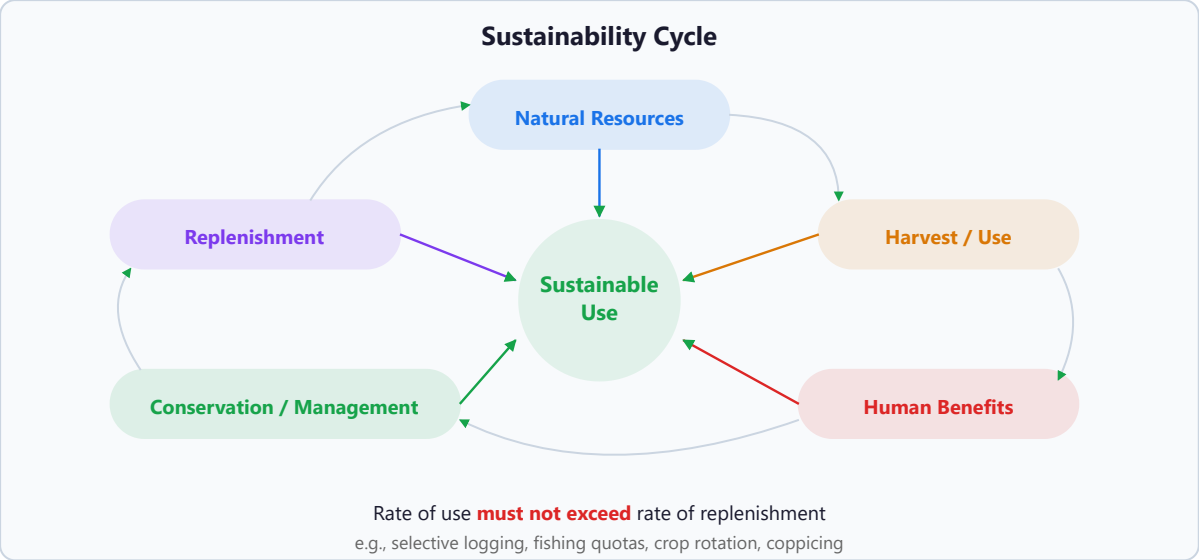
that allows them to be replenished naturally.



Sustainable practices include:

- **Sustainable forestry:** Selective logging, replanting, coppicing, and long rotation periods to maintain forest cover while providing timber.
- **Sustainable fishing:** Quotas, mesh size regulations, closed seasons, marine protected areas to allow fish stocks to recover.
- **Crop rotation:** Growing different crops in sequence to maintain soil fertility, break pest cycles, and reduce the need for chemical inputs.
- **Agroforestry:** Combining trees with crops or livestock to improve productivity, biodiversity, and soil quality.

4.5 Conservation of Biodiversity

Threats to Biodiversity

The main threats to global biodiversity include:

- **Habitat destruction:** Deforestation, urbanisation, agricultural expansion. The single greatest threat to biodiversity worldwide.
- **Habitat fragmentation:** Breaking continuous habitats into smaller, isolated patches, reducing gene flow between populations and making them more vulnerable to extinction.

- **Overexploitation:** Overhunting, overfishing, overcollection (e.g., of wild plants for traditional medicine).
- **Pollution:** Chemical pollution, eutrophication, plastic pollution, light and noise pollution.
- **Climate change:** Changing temperature and rainfall patterns, rising sea levels, ocean acidification.
- **Invasive species:** Non-native species that outcompete, prey on, or bring diseases to native species.

Conservation Strategies

In-situ conservation: Protecting species in their natural habitat. Examples include national parks, nature reserves, Sites of Special Scientific Interest (SSSIs), wildlife corridors, and marine protected areas. This maintains natural ecological interactions and allows evolution to continue.

Ex-situ conservation: Protecting species outside their natural habitat. Examples include zoos, botanic gardens, seed banks (e.g., Svalbard Global Seed Vault), captive breeding programmes, and frozen zoos (storing genetic material). These are important for critically endangered species but cannot conserve entire ecosystems.

> **Exam Tip:** In essay questions about conservation, discuss reasons for conserving biodiversity (ecological, economic, ethical, aesthetic), the advantages and limitations of both in-situ and ex-situ approaches, and suggest how the two can work together (e.g., captive breeding followed by reintroduction to protected habitats).

International Cooperation

Biodiversity conservation requires international agreements because species and ecosystems cross national boundaries. Key agreements include the Convention on Biological Diversity (CBD), CITES (Convention on International Trade in Endangered Species), the Rio Earth Summit targets, and national biodiversity action plans. However, enforcement can be difficult, and economic pressures often conflict with conservation goals in developing nations.

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